TEXORA · LECTURE SHEET

Doubling Machines: Up-twisters and Two-for-one Twister

Two or more singles yarns are wound together and then twisted into a plied yarn that is stronger, evener and more balanced than any of its parts. This sheet covers why doubling is done, how the up-twister works, and how the two-for-one twister inserts two twists for every turn of its spindle.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 8 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Purpose of doubling and twisting (plying)

Doubling means combining two or more yarns into one strand. It is done in two stages that must not be confused: *winding (assembly winding)* places the required number of singles yarns side by side on one package under equal tension, without twist; *twisting (plying)* then inserts ply twist, normally in the opposite direction to the singles twist, to lock the strands into a single plied (folded) yarn.

Plying is carried out because a plied yarn is better than a singles yarn of the same resultant count in almost every property that matters for demanding uses:

- **Strength.** A weak place in one strand is bridged by the others, so the plied yarn is stronger than a singles yarn of equal resultant tex, and much more consistent.
- **Evenness.** Thick and thin places average out when strands combine, so irregularity falls.
- **Twist balance.** Singles yarn is lively — it snarls when slack. Plying with opposite-direction twist cancels most of that torque, giving a balanced yarn that lies flat in fabric and does not skew knitted garments or form snarls in sewing.
- **Abrasion resistance, roundness and effects.** The plied structure is rounder, more compact and more abrasion-resistant, presenting several strands to wear — important for sewing threads, cords, ropes and warps. Strands of different count, colour, fibre or twist can also be combined deliberately — the basis of crepe, corded and fancy yarns.

Plying costs an extra winding and twisting process, so it is used where its gains are worth paying for — sewing threads, quality warps, non-skewing knitting yarns, industrial yarns and cords.

2. The up-twister

2.1 Working principle

The up-twister is the traditional ply-twisting machine, developed from the ring frame. The assembly-wound package of doubled yarn sits on a creel or spindle and the yarn is drawn *upwards* — hence the name — through a guide, around a rotating flyer or through a ring-and-traveller system, and onto a take-up package. Each revolution of the flyer or traveller inserts **one turn of twist** into the length of yarn between the feed and the twisting point:

Twist (turns/m) = twisting-element speed (rpm) / delivery speed (m/min) — one twist per revolution

2.2 Working sequence

1. The doubled yarn is unwound from the supply package under light, even tension, and passes through a guide and tensioner.
2. The tensioner keeps the strands together: unequal strand tension here becomes a permanent defect — a tight strand spiralling around a slack one (the "corkscrew" fault).
3. The rotating flyer (or ring traveller) inserts twist as the yarn travels to the take-up package.
4. Delivery rollers deliver yarn at a set speed; flyer speed ÷ delivery speed sets the ply twist.
5. The twisted yarn is wound onto the take-up package in a form suitable for the next process.

Up-twisters are simple, robust and flexible — twist, count and ply number change easily — which keeps them in use for small lots, coarse counts, fancy yarns and heavy industrial plies. Their weakness is speed: package size is limited and energy use per twist is high, so production per spindle is low by modern standards.

3. The two-for-one (TFO) twister

3.1 The central idea

In the two-for-one twister the supply package does **not** rotate: it sits stationary inside a protective pot, on a hollow spindle. The yarn leaves the package, passes through the hollow spindle axis, exits at the spindle base and balloons around the stationary package before rising to the delivery rollers and take-up package. Every spindle revolution twists the yarn **twice**:

- **First twist** — in the length of yarn between the package and the spindle exit, as the yarn is drawn through the rotating hollow spindle.
- **Second twist** — in the balloon length, between the spindle exit and the guide above the pot, as the balloon revolves once around the package.

Hence the name: **two twists for one spindle revolution**. Twist is therefore:

Twist (turns/m) = 2 × spindle speed (rpm) / delivery speed (m/min)

3.2 Yarn path, step by step

1. Assembly-wound supply package placed over the hollow spindle, inside the stationary pot.
2. Yarn drawn from the package through a tension device inside the spindle, which controls balloon tension.
3. Yarn passes through the hollow spindle along its axis and out through the rotating yarn eye / disc at the spindle base.
4. The yarn forms a balloon around the pot as the spindle rotates — one revolution of the balloon per spindle revolution.
5. Yarn rises through an overhead guide eye, which sets the balloon height.
6. Delivery (take-off) rollers draw the yarn at the set delivery speed.
7. The twisted yarn is wound onto a large cross-wound take-up package, ready for warping, knitting or dyeing without rewinding.

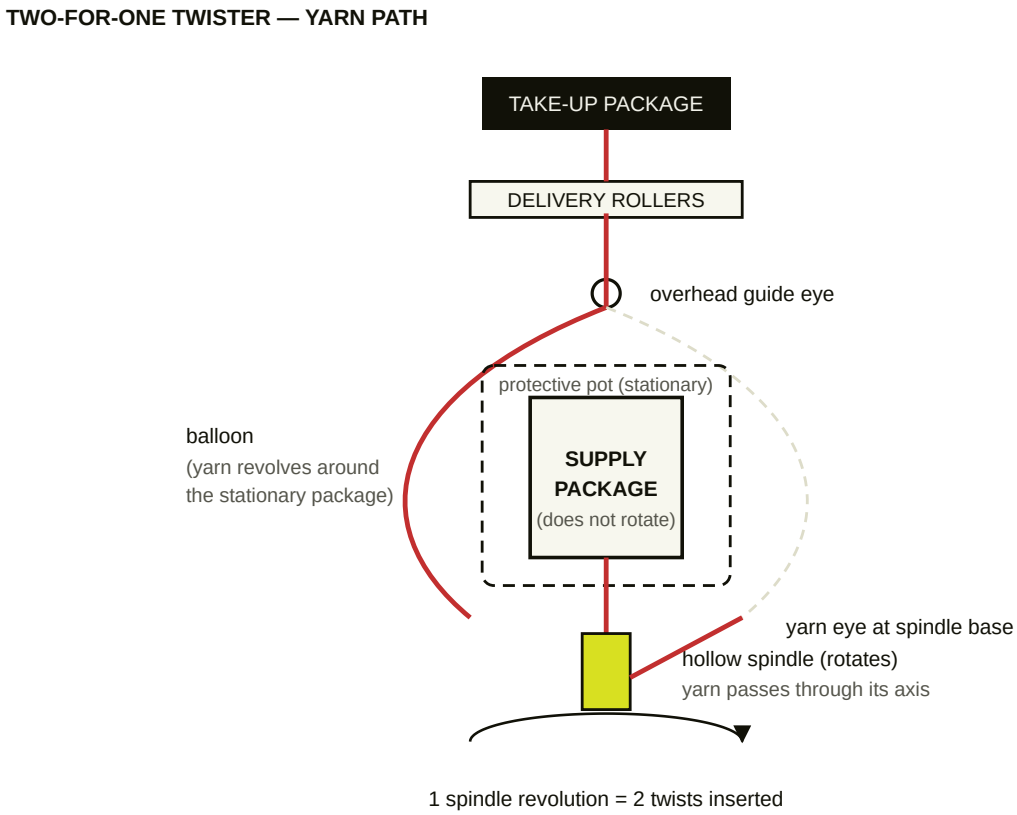


Figure 1 — Two-for-one twister. The yarn travels from the stationary package through the hollow spindle, balloons once around the package, and rises through the guide eye to the delivery rollers and take-up package.

3.3 Advantages and limits

- **Double twist per revolution** — for a given spindle speed the machine inserts twice the twist, or inserts a given twist at half the delivery limitation of a one-twist machine; production per spindle is high.
- **Large packages both ways.** Supply and take-up packages are both large; the take-up needs no further winding — fewer knots and joins, longer lengths for warping and knitting.
- **Energy and quality.** Only the small spindle and balloon rotate, not a heavy package, so energy per kilogram is lower than on up-twisters, and twist is inserted gently with good evenness.
- **Limits.** Balloon tension rises sharply with speed and package size, so very coarse yarns, very high twists and some fancy yarns are difficult; the machine costs more and is more complex to thread and maintain, and balloon contact must be controlled to avoid abrasion and snarling.

4. Up-twister versus two-for-one twister

FEATURE	UP-TWISTER	TWO-FOR-ONE (TFO) TWISTER
Twists per revolution	1	2
Supply package	Rotates / is circled by flyer	Stationary, inside a pot on a hollow spindle
Package size	Small to moderate	Large supply and large take-up
Production per spindle	Low	High
Energy per kg	High	Lower
Flexibility / small lots	Very flexible, simple to set	Best for longer runs of standard plies
Coarse / fancy / very high twist	Well suited	Limited by balloon tension
Typical role today	Speciality, coarse and fancy plying	Main production ply-twisting machine

WORKED EXAMPLE — TFO TWIST AND PRODUCTION

Spindle speed = 10,000 rpm. A TFO inserts two twists per revolution, so twists inserted = 2 × 10,000 = **20,000 twists per minute**.

Delivery speed = 45 m/min. Twist = 20,000 / 45 = **444 turns/m** (rounded).

Production length per spindle in 8 h = 45 × 60 × 8 = **21,600 m**.

For a resultant plied yarn of 40 tex: mass = 21,600 × 40 / 1,000,000 = **0.864 kg per spindle per 8 h**.

> An up-twister at the same speeds would give only 10,000 / 45 = 222 turns/m — the TFO doubles it.

Tension control is the quality point on both machines. Assembly winding must lay every strand at equal length and tension, and twisting tension must be steady: tension variation becomes twist variation, which shows in fabric as streaks, uneven dyeing and local skew. Ply twist is chosen to balance the singles twist — check by letting a slack length hang in a loop: a balanced yarn does not rotate or snarl.

Glossary

Doubling — combining two or more yarns into one strand. **Plying / folding** — twisting doubled strands together into a plied yarn. **Assembly winding** — winding the strands together on one package, without twist, before plying. **Up-twister** — ply-twisting machine inserting one twist per flyer/traveller revolution. **Two-for-one twister** — machine inserting two twists per spindle revolution, with a stationary package on a hollow spindle. **Balloon** — the revolving curve of yarn around the package between spindle exit and guide eye. **Corkscrew fault** — one strand spiralling around a slacker strand (unequal strand tension). **Resultant count** — the count/tex of the finished plied yarn.

Key points

• Doubling = assembly winding (no twist) followed by ply twisting (usually opposite in direction to the singles twist).
• Plying raises strength, evenness, abrasion resistance and roundness, and balances singles torque.
• Up-twister: one twist per revolution; simple and flexible, but slow with small packages.
• TFO: yarn through a hollow spindle plus one balloon revolution = two twists per spindle revolution; stationary large package, large take-up, high production.
• TFO twist (turns/m) = 2 × spindle rpm / delivery (m/min); up-twister twist = spindle rpm / delivery.
• Equal strand tension before and during twisting is essential — unequal tension causes corkscrew and twist-variation faults.

Practice

1. A TFO twister runs at a spindle speed of 12,000 rpm with a delivery speed of 50 m/min. Calculate the ply twist in turns/m. Then calculate the production length per spindle in 8 h, and the mass produced if the resultant yarn is 36 tex.
2. State the two places where twist is inserted in a TFO twister, and explain why losing either one would halve the twist at unchanged machine settings.

3. A plied knitting yarn skews garments badly. Explain, using twist balance, the likely cause and the correction to make at the twisting machine.
4. Give two jobs for which you would still choose an up-twister over a TFO twister, with a reason for each.

Answer key

1. Twists inserted = 2 × 12,000 = 24,000 per minute. Twist = 24,000 / 50 = **480 turns/m**. Length in 8 h = 50 × 60 × 8 = **24,000 m**. Mass = 24,000 × 36 / 1,000,000 = **0.864 kg per spindle per 8 h**.
2. Twist is inserted (1) in the yarn length passing through the rotating hollow spindle, and (2) in the balloon length as the yarn revolves once around the package. Each contributes one twist per spindle revolution; if the yarn did not balloon (or did not pass through the rotating spindle), only one twist per revolution would remain, halving the total twist at the same speeds.
3. The plied yarn is torque-unbalanced: the ply twist does not cancel the singles twist, so the yarn tries to untwist in the fabric and the wales slant. Correction: adjust the ply-twist level (and confirm its direction is opposite to the singles twist) until a slack loop of the yarn hangs without rotating or snarling, then re-test the knitted fabric for skew.
4. Examples: (a) a short run of a fancy plied yarn — the up-twister is quicker and simpler to set for a small, frequently changed lot; (b) a very coarse or very highly twisted industrial ply — balloon tension on a TFO would be excessive or uncontrollable, while the up-twister handles it directly. Any two well-reasoned cases of this kind are acceptable.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib